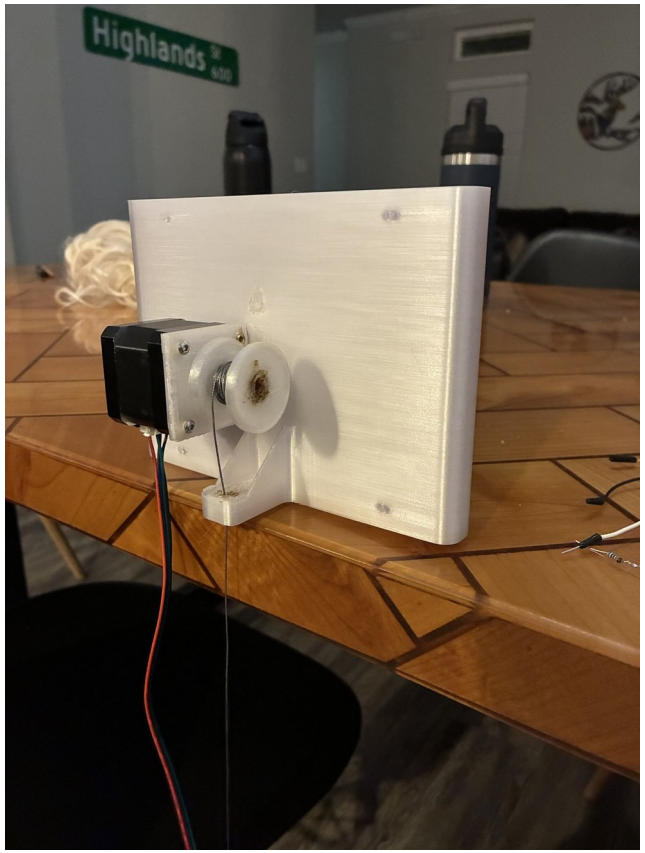
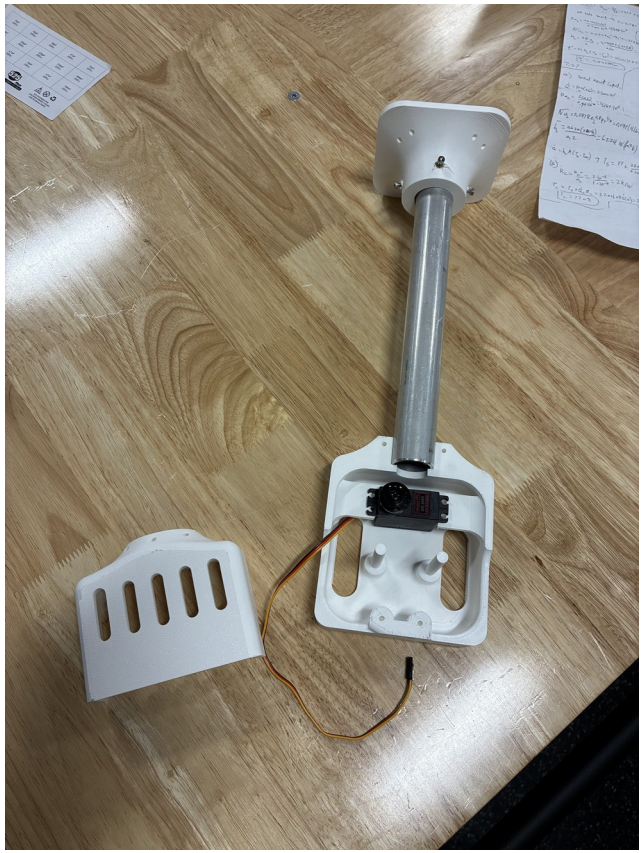
RoomCleaner

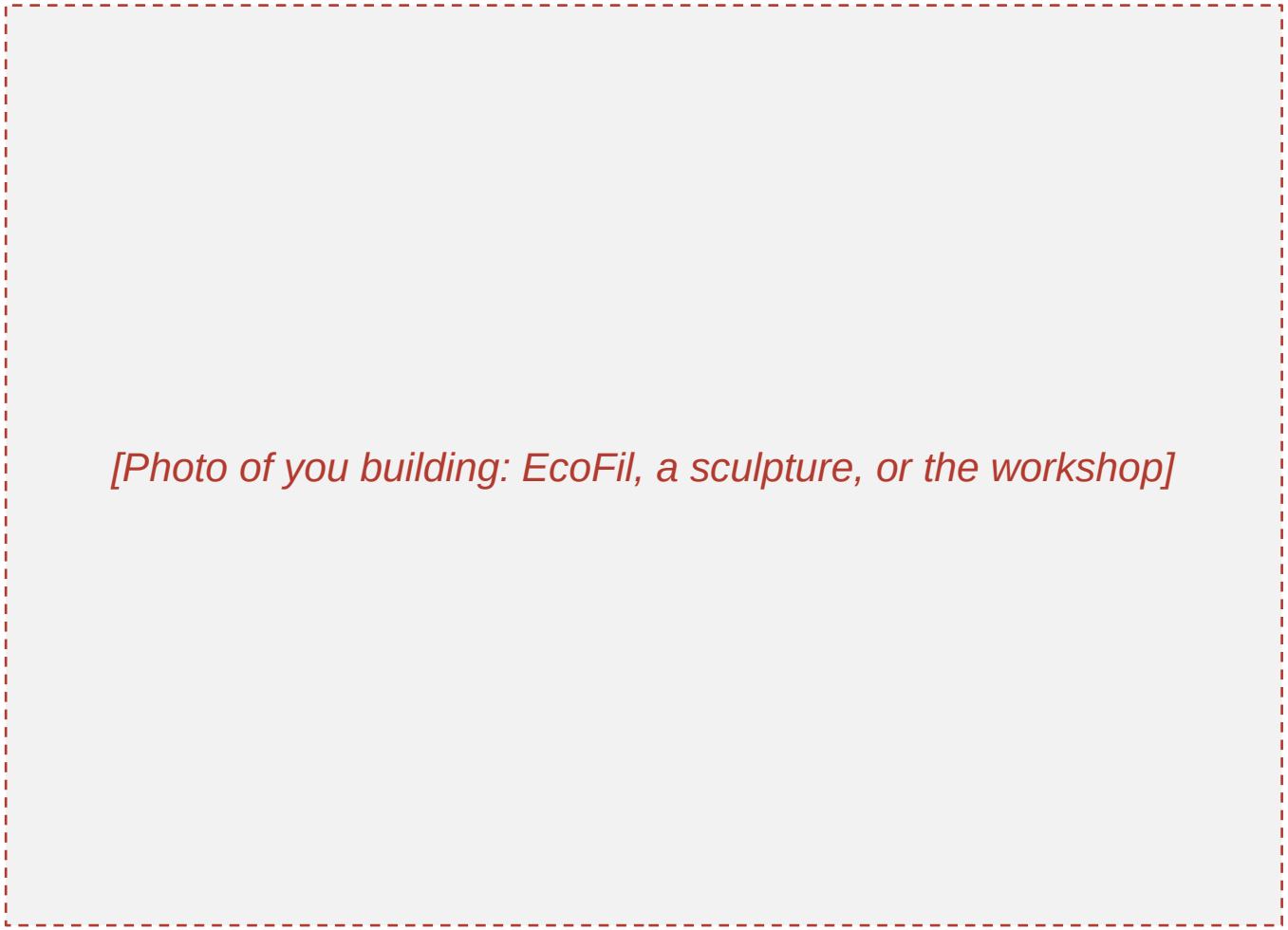
A cable-driven robot that picks laundry up off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026



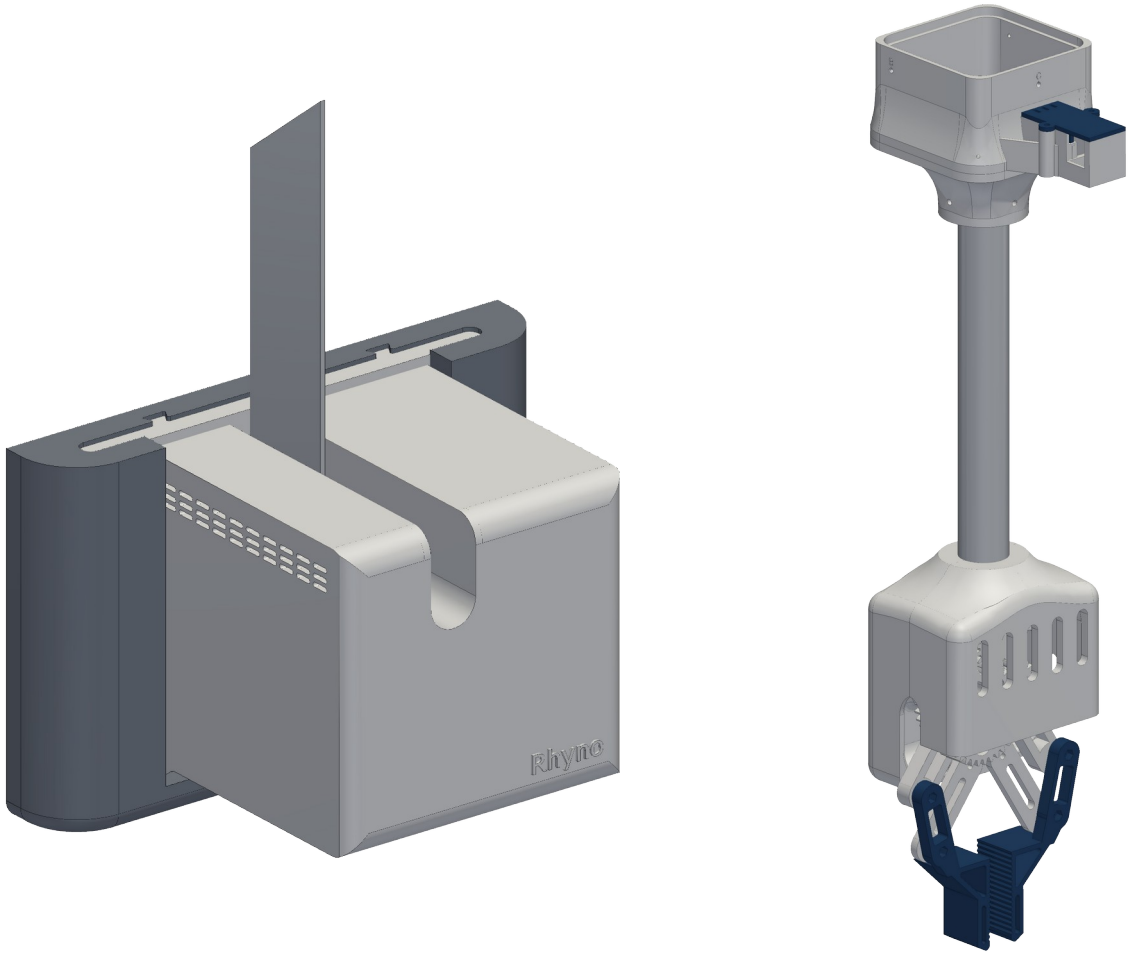


About Me



- Mechanical Engineering, Texas A&M University, *[graduation year]*
- Built EcoFil, a filament recycler: shredder, dryer and extruder
- Metal sculpture with fiber laser cutting and welding
- *[hobbies]*

Project Goal

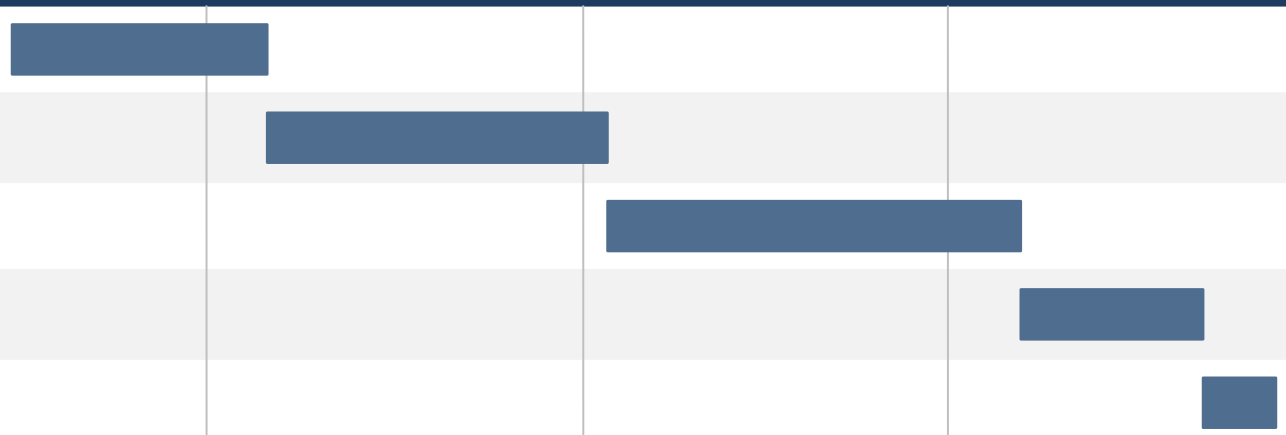


Final design (CAD): wall station and claw

- Pick up laundry anywhere in an 11 × 9 ft room and drop it in a hamper
- Design load: jeans, 0.9 kg, at 2 m/s²; first test with 1 lb
- 3D-printed PETG (polyethylene terephthalate glycol) on a hobby budget
- Success: one garment to the hamper with no human input

Timeline and Scope

Phase	Dates	July	August	September	October
1. Concept	Jul 16 – Aug 5				
2. Generation 1	Aug 6 – Sep 2				
3. Redesign	Sep 3 – Oct 6				
4. Final design	Oct 7 – Oct 21				
5. Build and test	Oct 22 – Oct 27				



- Solo project: requirements, calculations, all CAD (computer-aided design), printing, machining, wiring and *[software parts you wrote]*
- Purchased: stepper motors, drivers, controller boards, servo, camera, battery and encoder boards
- Garment detection uses YOLO-World, an open-source model

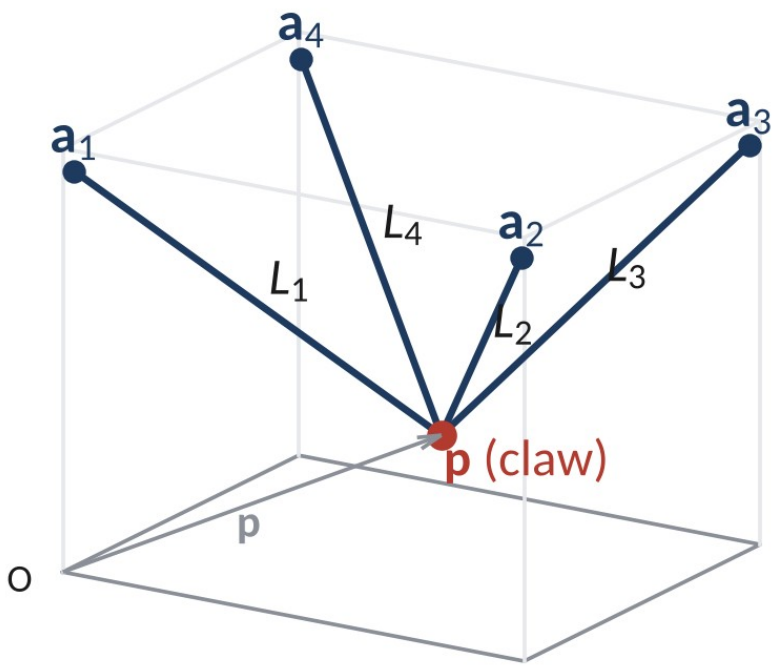
Phase 1

Concept

July 16 – August 5

Choose the architecture and set the limits

Why a Cable Robot



- Four ceiling winches hold one claw: the whole floor is reachable and nothing stands on it
- The four cable lengths set the claw position
- A laptop plans every move; a phone app starts, pauses and approves it

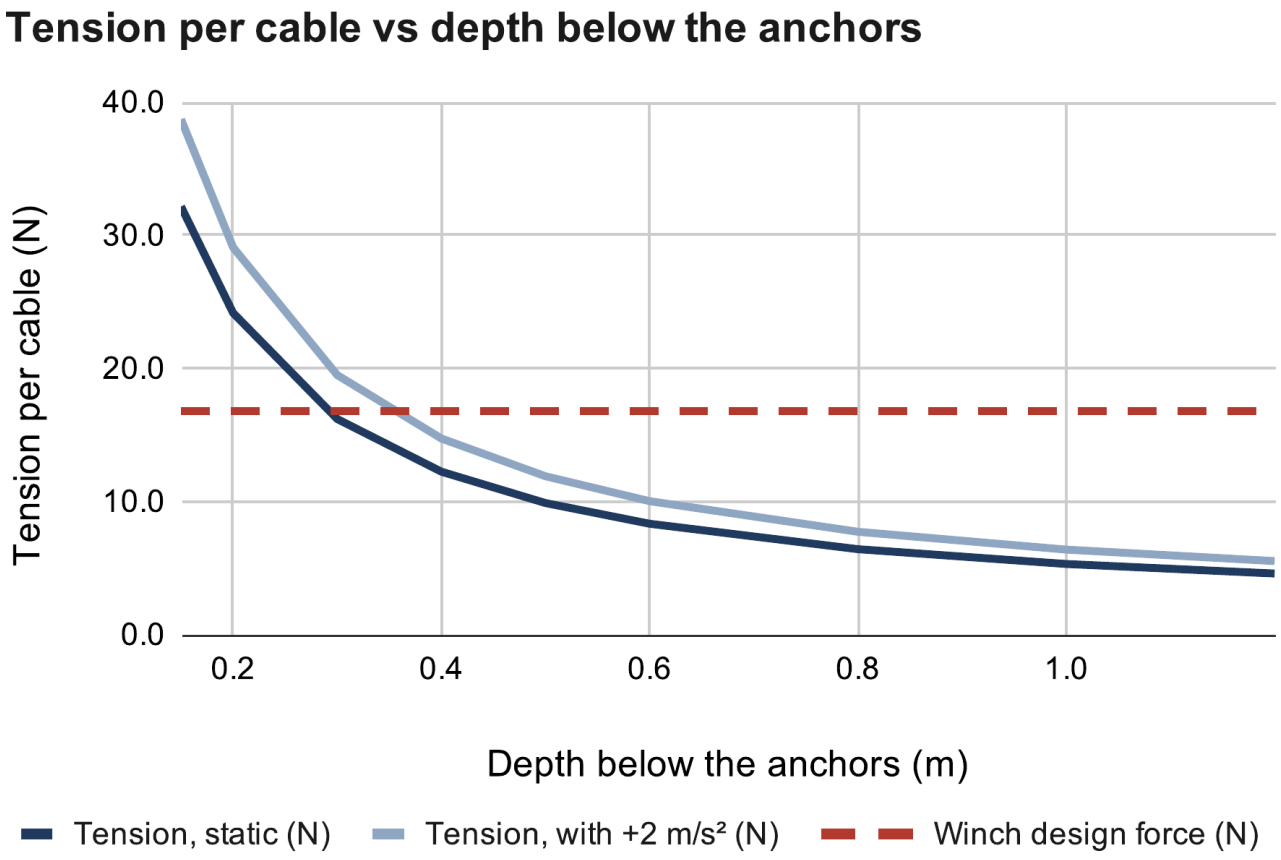
Cables run from the ceiling anchors a_1 to a_4 to the claw at p

Design Trade-offs

Design change	Benefit	Cost	Chosen setting
Higher motor current	41 % more pull at 1.5 A	Heat rises with current squared	1.5 A, vented shell and heat fin
Stronger claw	Gear safety factor 0.4 to 2.0	More claw mass, less payload	Module 1.5 gears
Larger room	More floor covered	Longer wires, more voltage drop	Only 12 V power runs to stations
Closer to the ceiling	More working volume	Cable tension 12 N to 32 N	Stay 0.4 m below the anchors
Smaller spool radius	More pull per amp	Slower line speed	15 mm core: 25–34 N pull
Larger safety margins	No skipped steps or wall contact	Less usable force and floor	Force ÷ 1.5; 0.2 m from walls

- A heavier claw needs more pull, more pull needs more current, and heat rises with the square of the current

Cable Tension Limit



Source: RoomCleaner engineering calculations, “Cable tension” (calculated)

- Tension rises sharply near the ceiling: 32 N at 0.15 m, 8.3 N at 0.6 m
- Operating limit set 0.4 m below the anchors
- Covers 2 m/s² acceleration within the 16.8 N winch limit

Tension per cable at 0.4 m

$T = W / (4 \sin \varphi)$

$= 8.87\ \text{N} / (4 \times 0.181) = \mathbf{12.2\ N}$

With 2 m/s²: **14.7 N** < 16.8 N

Phase 2

Generation 1

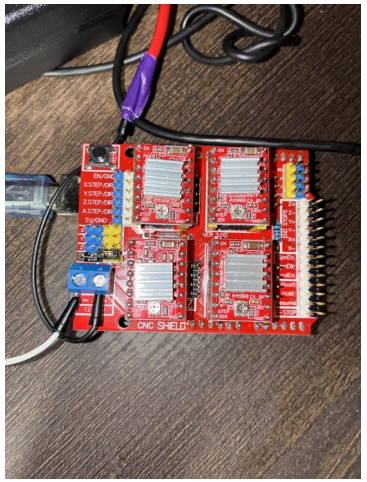
August 6 – September 2

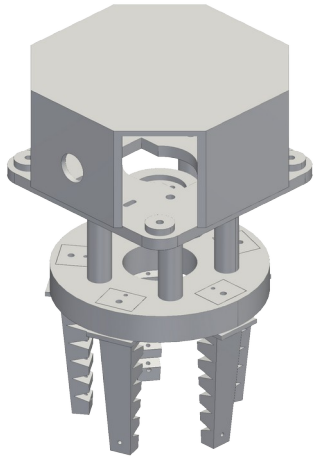
Prove the idea on the bench

Generation 1 Build





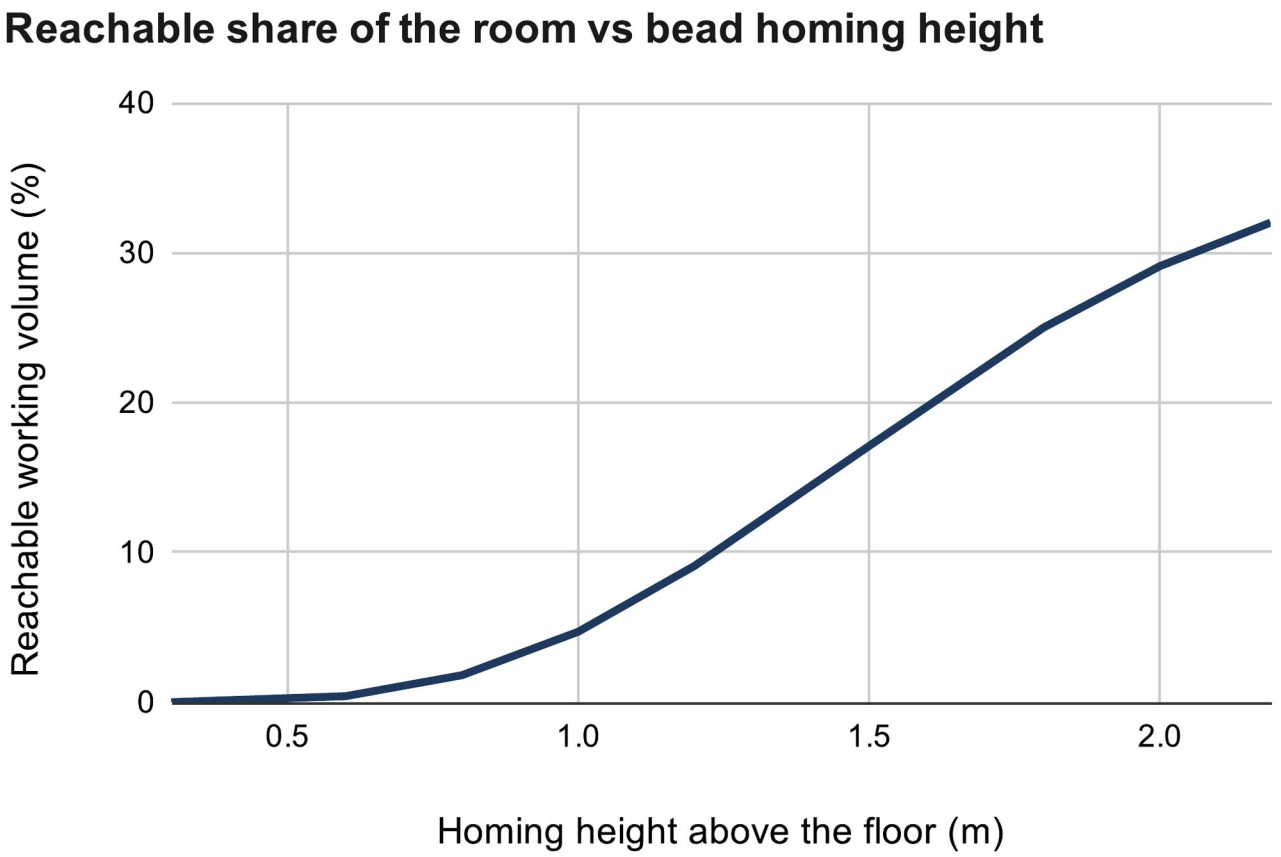




Bench lift test, corner plate, central driver board, finger claw (CAD)

- One-piece corner plates: motor, spool, pulley and home switch
- One central board drove all four motors
- Five flexible TPU (thermoplastic polyurethane) fingers on the claw
- A station lifted a test weight on the bench *[weight]*

What Generation 1 Taught Me



- Bead homing left only 5–29 % of the room reachable
- Adding a sensor at each winch meant 10 wires per corner
- Finger claw could not pinch fabric and was 4.6 times over its mass budget

Source: RoomCleaner engineering calculations, “Bead homing” (calculated)

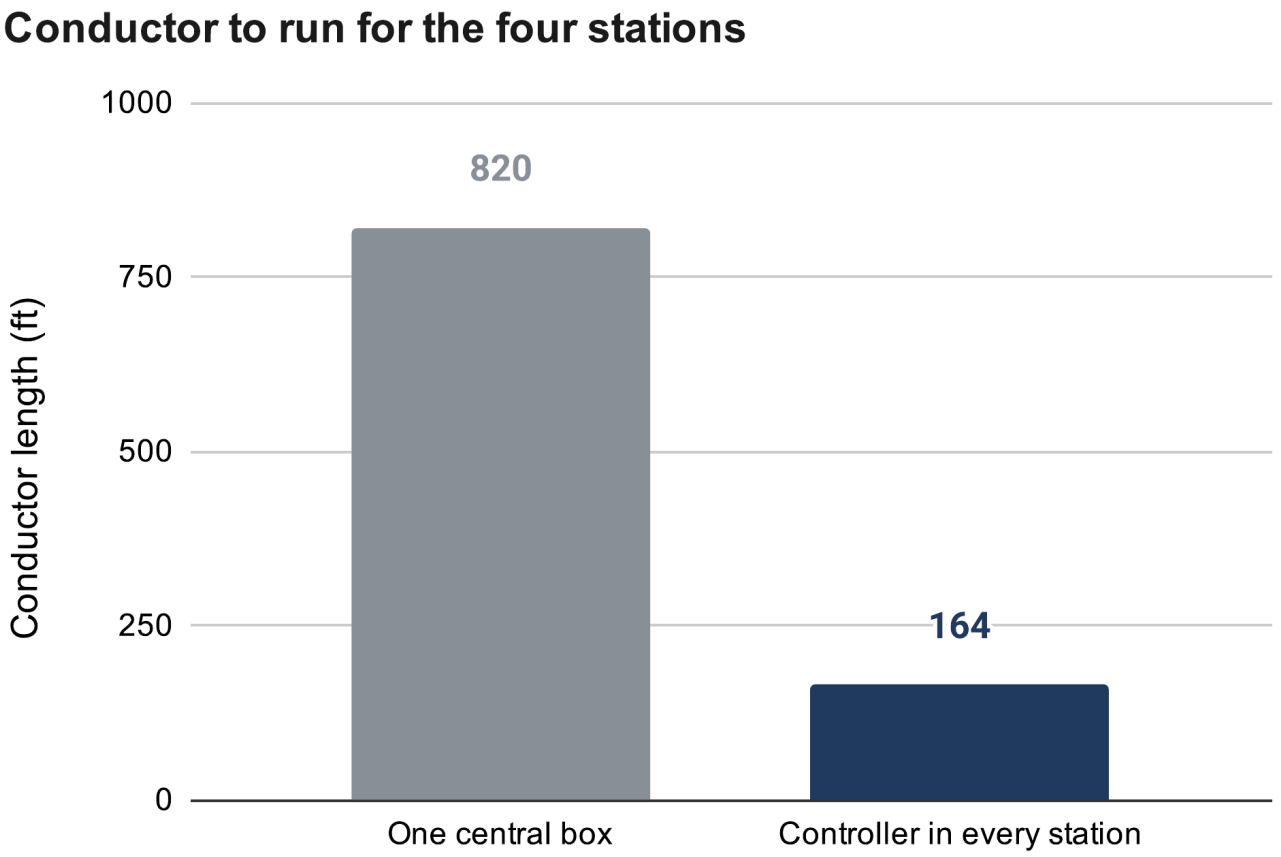
Phase 3

Redesign

September 3 – October 6

Fix what generation 1 exposed

One Controller per Winch



Source: RoomCleaner engineering calculations, “Wiring” (calculated)

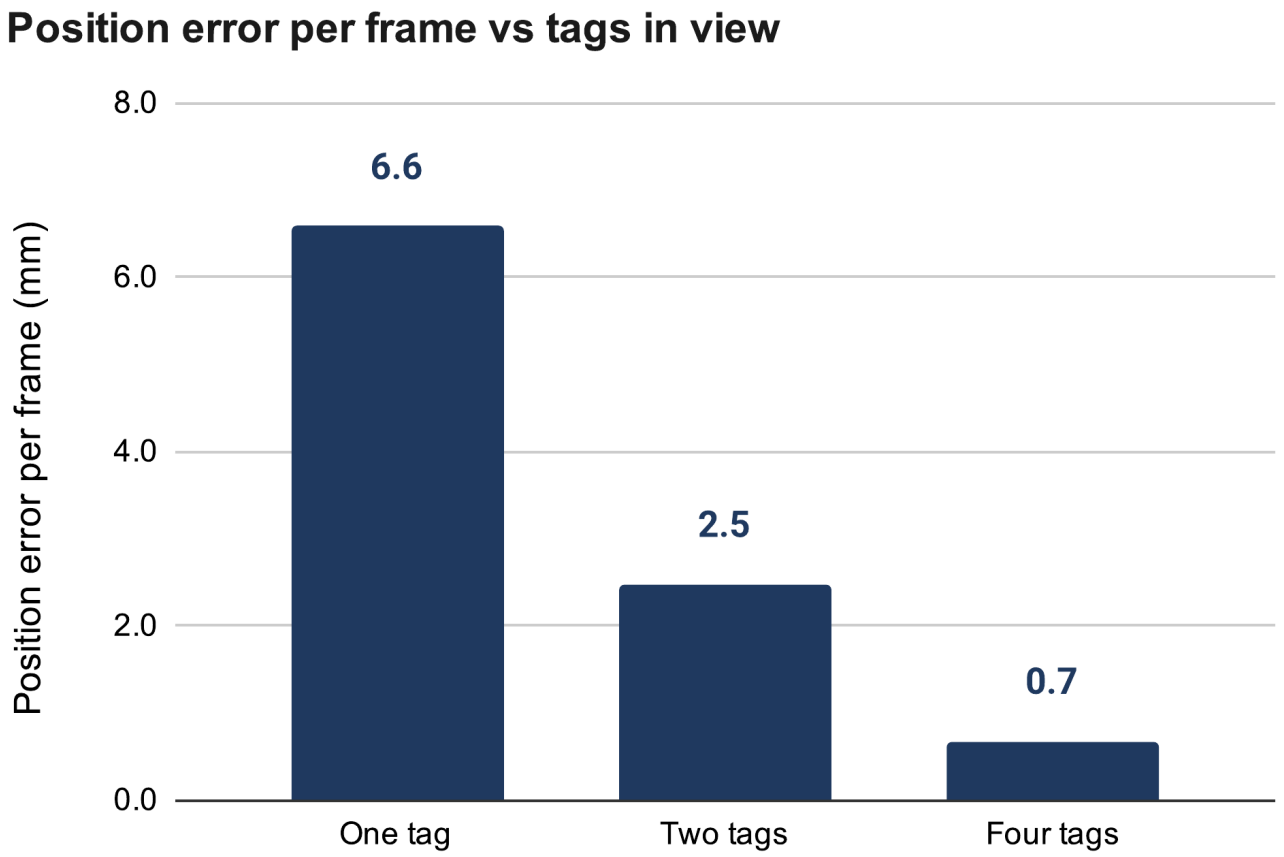
- A controller in each station instead of one central board
- Only a 12 V pair runs to each station: five times less wire
- Cost: synchronization moves into software

Wire to the four stations

Central board: 10 × 82 ft = **820 ft**

Per station: 2 × 82 ft = **164 ft**

Camera Calibration Replaces Homing

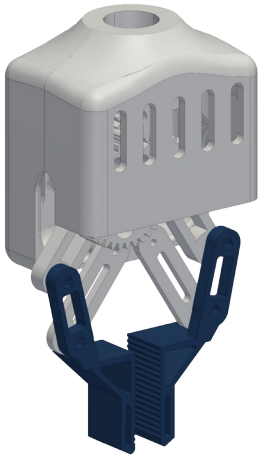


- The claw camera reads a printed floor board from several poses
- A fit finds the anchor positions and cable lengths: no tape measure
- Position error: 6.6 mm with one tag in view, 0.7 mm with four
- Whole room reachable; the beads stay as end stops

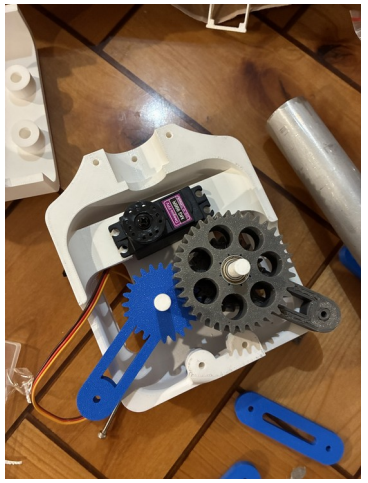
Source: RoomCleaner engineering calculations, “Floor board” (calculated)

Geared Four-Bar Claw









v6 clamp, final four-bar (CAD), printed claw body and gear train

- Parallel jaws pinch thin fabric; 53 mm opening
- Camera moved into the claw to see each grasp
- Stall check moved the gears from module 1 to 1.5

Gear stress at servo stall

$\sigma = F_t / (b\ m\ Y) =$ **11.0 MPa**

$SF = 22.5\ MPa / 11.0\ MPa =$ **2.05**

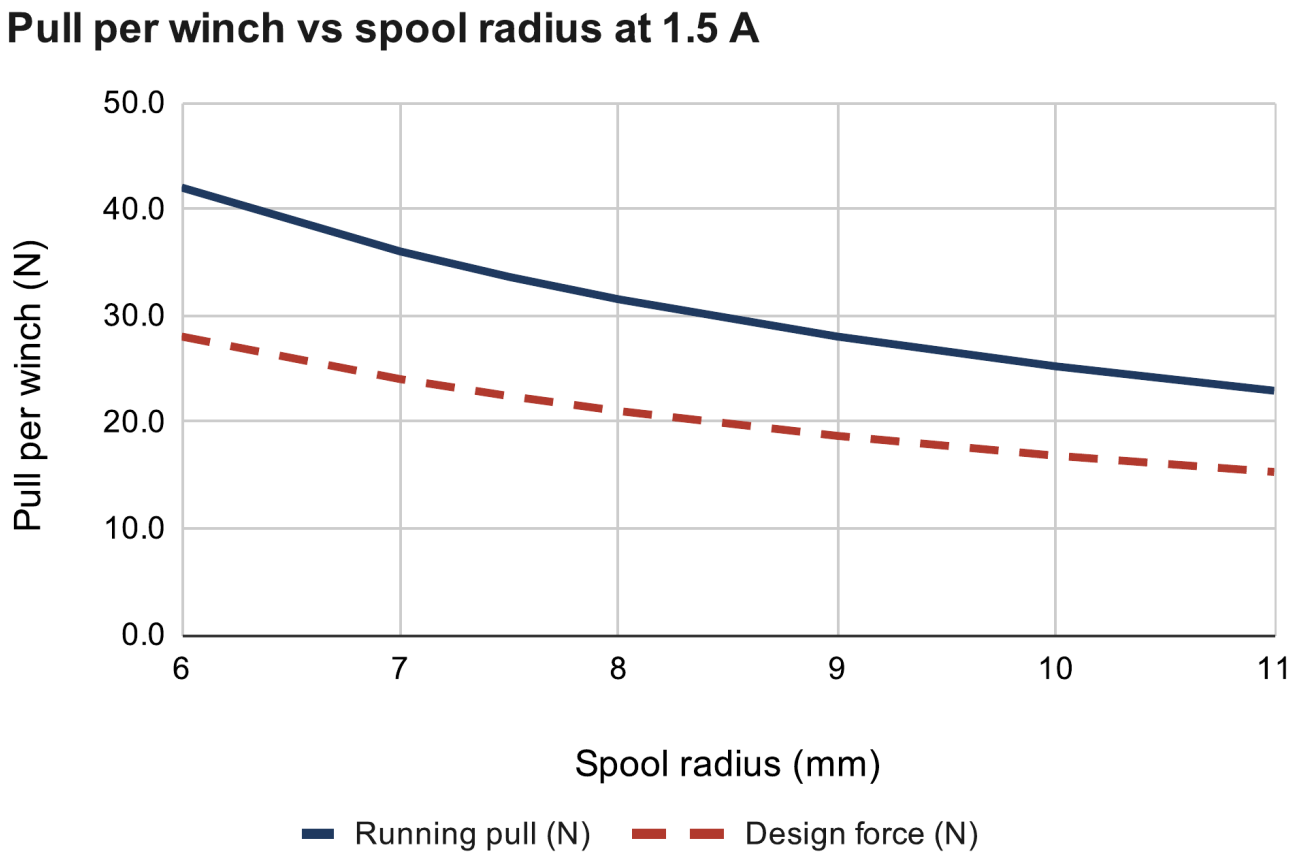
Phase 4

Final Design

October 7 – October 21

Size the winch, the cooling and the weight in SolidWorks

Winch Pull at 1.5 A



Source: RoomCleaner engineering calculations, “Winch pull” (calculated)

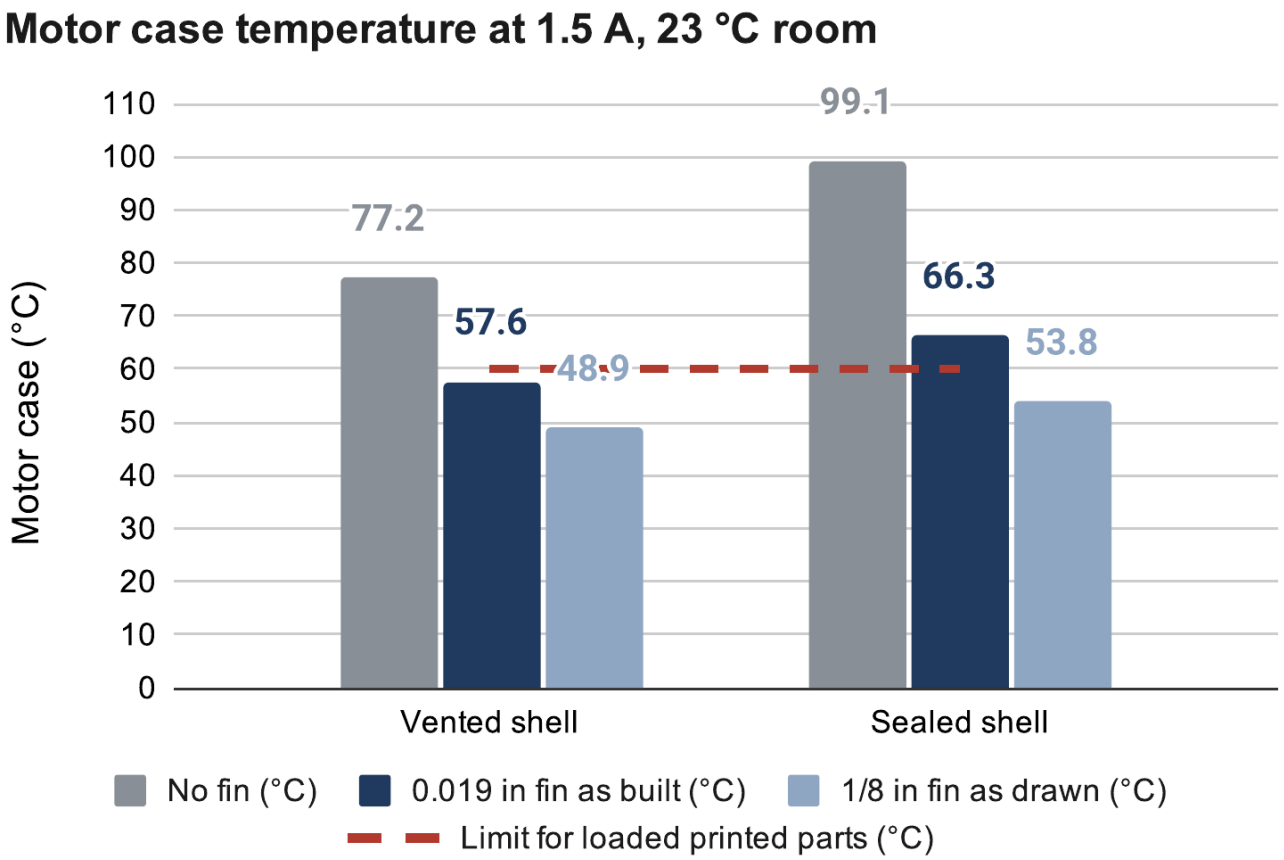
- NEMA 17 (National Electrical Manufacturers Association) stepper on a printed spool
- Driver current raised from 1.06 A to the rated 1.5 A: 41 % more pull
- 34 N with an empty spool, 25 N when full

Winch design force

$T = \tau / r = 0.252\ \text{N·m} / 0.010\ \text{m} = \mathbf{25.2\ N}$

$F_{design} = 25.2\ \text{N} / 1.5 = \mathbf{16.8\ N}$

Motor Cooling



Source: RoomCleaner engineering calculations, “Motor temperature” (calculated)

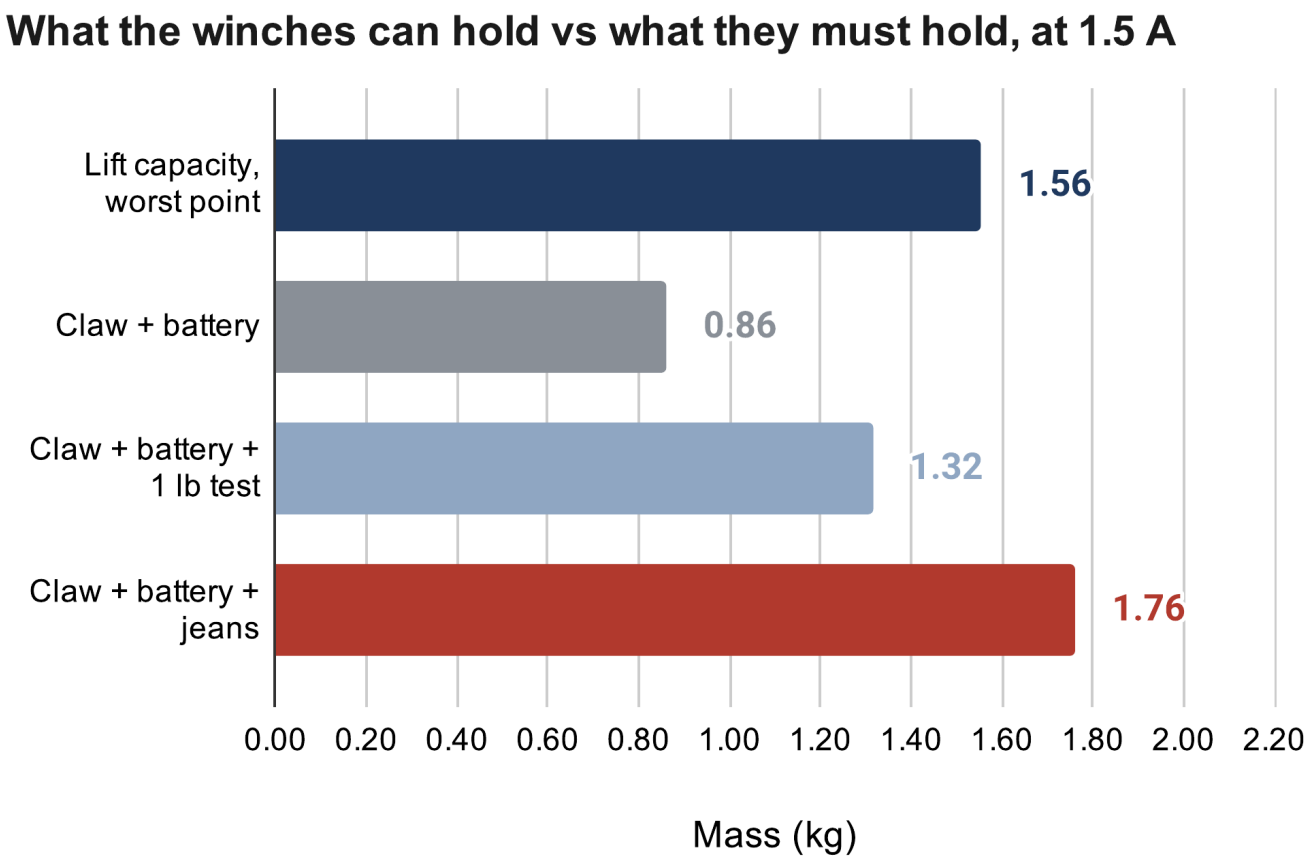
- 1.5 A doubles motor heat to 4.05 W
- PETG creeps above 60 °C, so the shell is vented and finned
- Motor at 45–49 °C with the 1/8 in fin, 56–59 °C with the 0.019 in prototype

Motor heat, $P = I^2 R$

At 1.06 A: $(1.06\ \text{A})^2 \times 1.8\ \Omega = 2.0\ \text{W}$

At 1.5 A: $(1.5\ \text{A})^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$

Weight Budget: Open Issue



Source: RoomCleaner engineering calculations, “Lift budget” (calculated)

- Lift capacity at 1.5 A: 1.56 kg
- Claw with battery estimated at 0.86 kg *[weighed]*
- 1 lb test fits; jeans are 0.2 kg over
- Next: a lighter claw or a gear reduction at each winch

Lift capacity

$m_{max} = F_{design} / (k\,g)$

$= 16.8 / (1.101 \times 9.81) =$ **1.56 kg**

Needed with jeans: **1.76 kg**

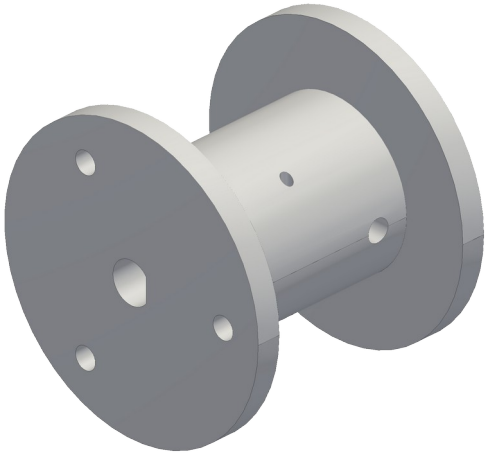
Phase 5

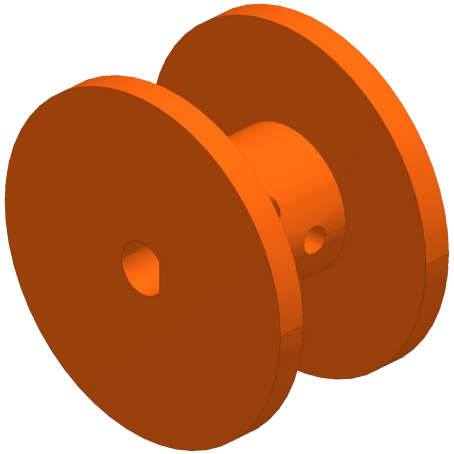
Build and Test

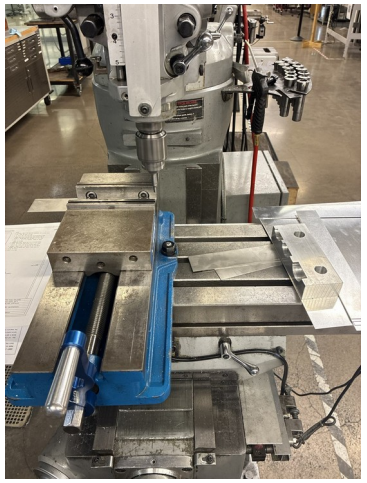
October 22 – October 27

Build the stations and run the first system test

Printing and Machining





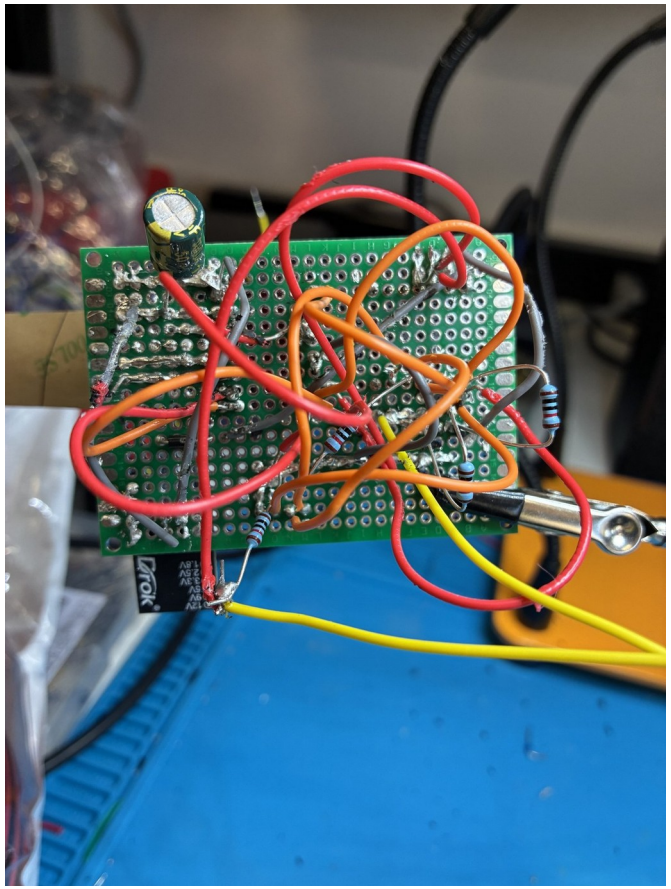




Spool v2 (bore too tight) and v3, fin mill setup, finished fin

- Printed holes come out at least 0.2 mm undersize: the spool took three prints
- Fit coupons are now printed before full parts
- Fin holes drilled in one mill setup to ±0.1 mm

Electronics Bring-Up





- Driver stays off at boot, so a reset cannot move a motor
- First motor failed to turn: logic rail at 1.14 V, not 3.3 V
- Supply rails are now checked with a meter before any code

Soldered station board and bench bring-up

First System Test

[Photo or video from the three-station test]

Check	Pass limit	Result
Self-check	All stations answer; slack taken up	*[fill in]*
Calibration	Held-out poses inside the margin	*[fill in]*
Short move, 1 lb	300 mm at 50 mm/s, ≤ 2 mm per cable	*[fill in]*
Pickup	One garment into the hamper	*[fill in]*

- Three stations: the fewest cables that fully hold the claw
- Self-check, calibration, short move, then pickup
- 1 lb first, then one garment; pass limits set before the run

Thermal Verification

[SolidWorks temperature plot: motor, fin, screws and mount]

Method	0.019 in fin	1/8 in fin
Hand calculation	56–59 °C	45–49 °C
SolidWorks	*[fill in]*	*[fill in]*
Hold test	*[fill in]*	*[fill in]*

- PETG mount must stay below 60 °C
- Hand calculation, SolidWorks study, then a 90-minute hold test
- The PETG runs at about the motor temperature

Results Against Targets

Target	Goal	Current	Status
Lift capacity	1.76 kg with jeans	1.56 kg (calculated)	**Not met**
Claw mass	0.45 kg	0.86 kg with battery; *[weighed]*	**Not met**
Motor temperature	Below 60 °C	45–49 °C with the 1/8 in fin	Met in design
Working area	Full 11 × 9 ft floor	All but 0.2 m along the walls	Met in design
Wiring	One cable per station	One 12 V pair; 0.51 V drop	Met
Accuracy	2 mm per cable	0.7 mm (calculated)	To be tested
Speed	50 mm/s at 2 m/s²	Not yet run	To be tested
Cost	Hobby budget	*[total cost]*	*[fill in]*

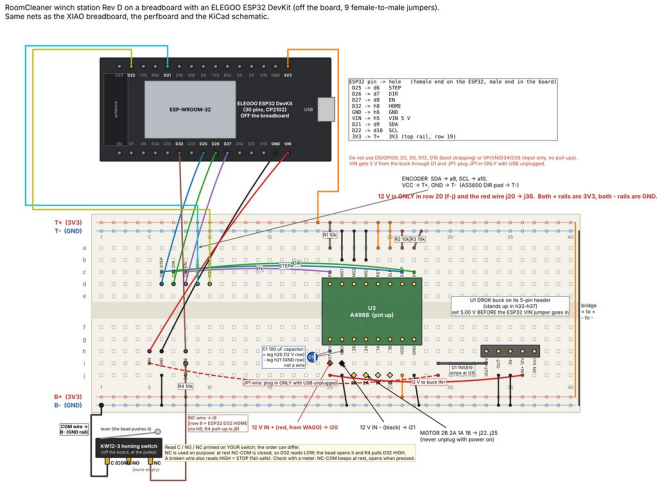
Good and Poor Decisions

Good decision	Result
One controller per winch	Five times less wire; each station tested alone
Camera calibration instead of bead homing	Whole room reachable
Gear strength checked at servo stall	Failure caught in CAD, before printing

Poor decision	Result
Mass budget set late	0.2 kg over capacity with jeans
Bead homing as the first plan	Under a third of the room reachable
Several stations built before one was proven	Every fix repeated on each station *[confirm]*

Thorough vs. Scrappy

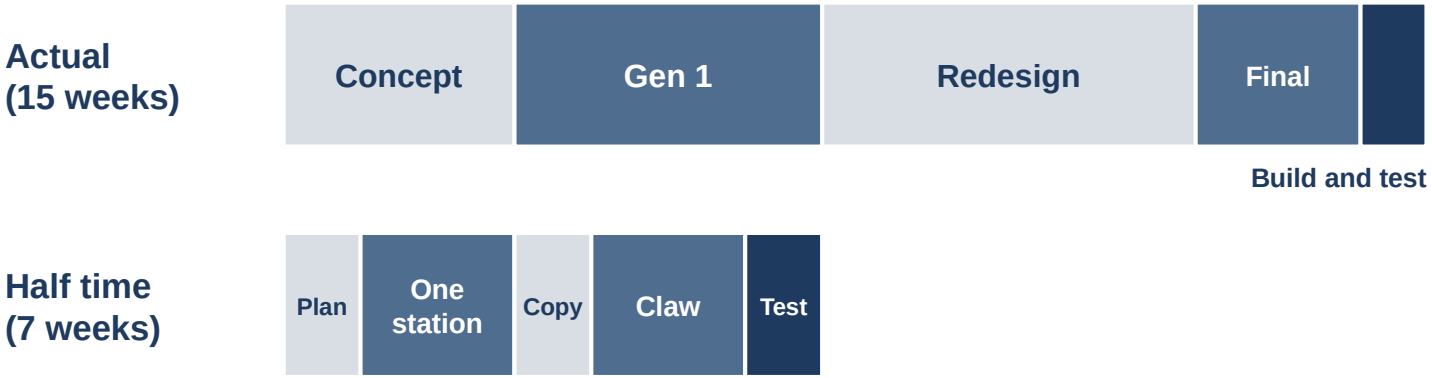




- Thorough where a mistake is expensive or unsafe
  - Heat fin: one mill setup, ±0.1 mm
  - Gears: strength checked before printing
- Scrappy where it is cheap to undo
  - Failed controller replaced by a jumpered ESP32 board
  - Floor board printed on paper

Thorough: fin mill setup. Scrappy: jumpered ESP32 controller board.

Lessons Learned



Same time scale. Half-time plan: prove one station, then copy it.

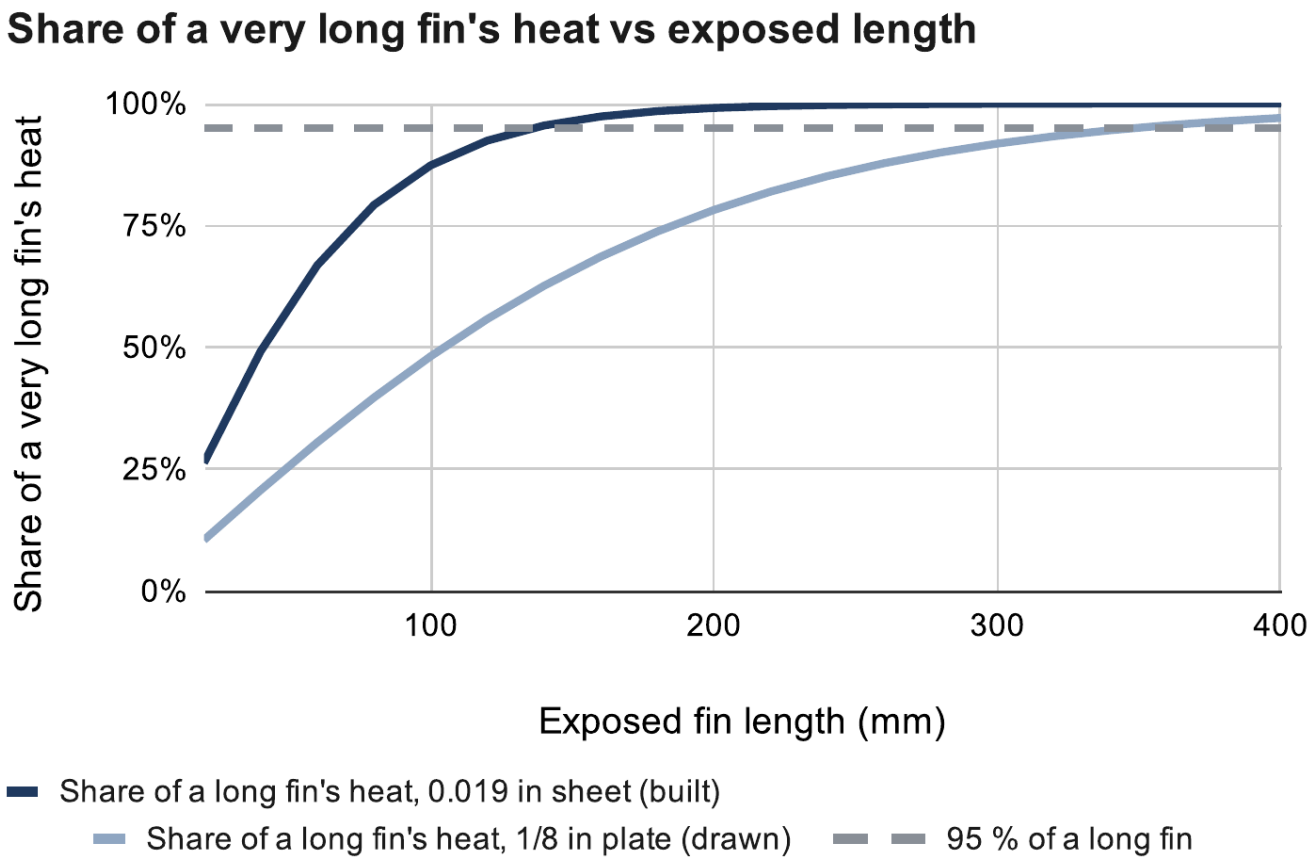
- Set the mass budget first, from the measured winch pull
- Prove one station before building the rest
- Print fit coupons before full parts
- In half the time: freeze requirements in week 1 and order long-lead parts on day 1

Questions

Jasper Buntinx | jasper_buntinx@tamu.edu

Appendix

Heat Fin Length



- 0.019 in prototype reaches 95 % of its capacity by 136 mm
- 1/8 in design keeps gaining to 349 mm
- Built at 200 mm, limited by ceiling clearance

Fin length and efficiency

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

$\eta = \tanh(mL) / (mL) =$ **0.74** (1/8 in)

Source: RoomCleaner engineering calculations, “Heat fin” (calculated)

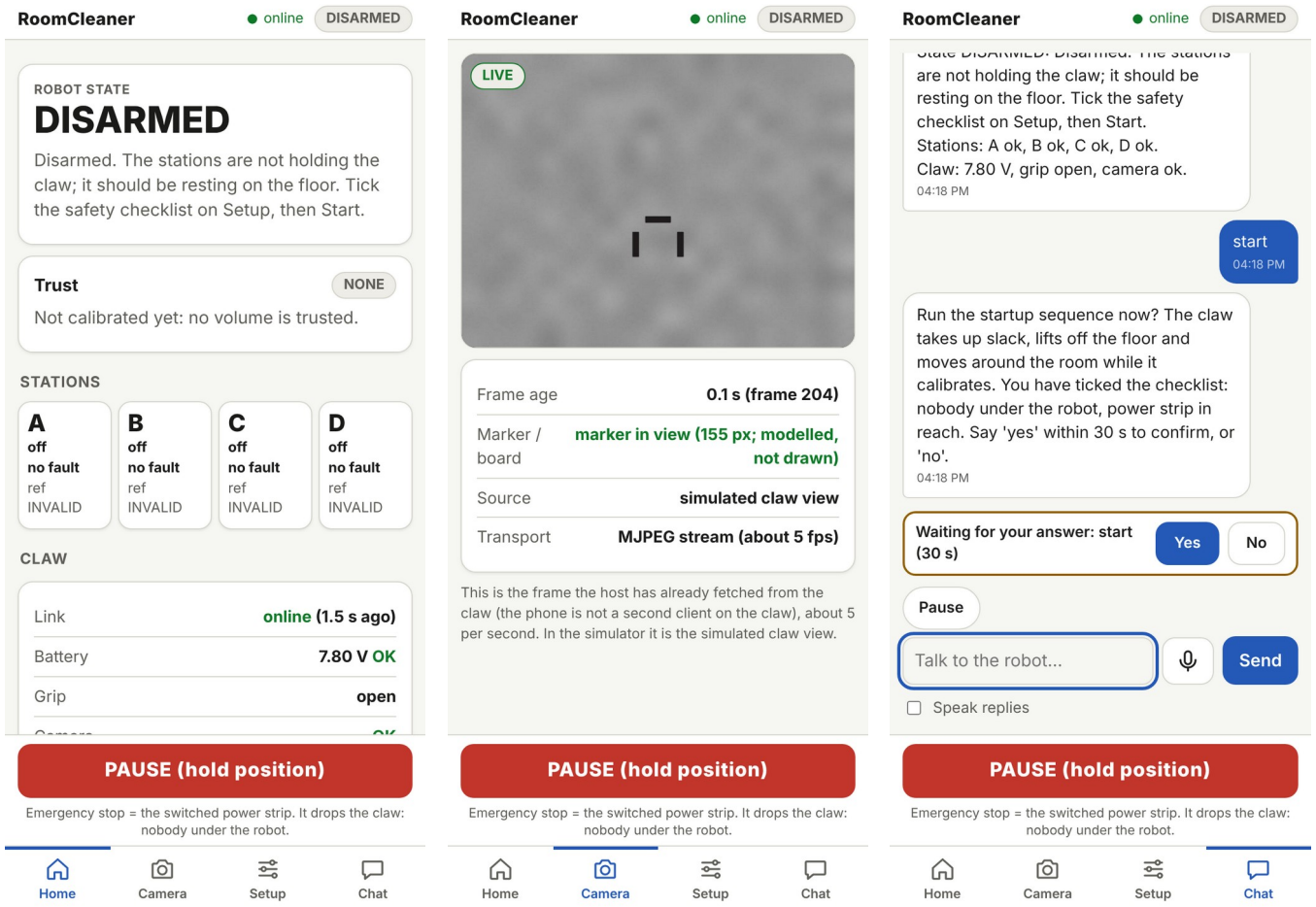
Fin Sizing for 4 W

Painted 6061, 44 mm wide	Mass	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	None (2.0 W max)
1/16 in	47 g	None (3.6 W max)
0.090 in	68 g	210 mm or none
1/8 in (design)	94 g	145–185 mm
3/16 in	141 g	125–140 mm
1/4 in	189 g	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

- If the fin alone had to carry all 4.05 W
- 0.019 in sheet: not possible at any length
- 1/8 in plate: 145–185 mm

Operating Sequence



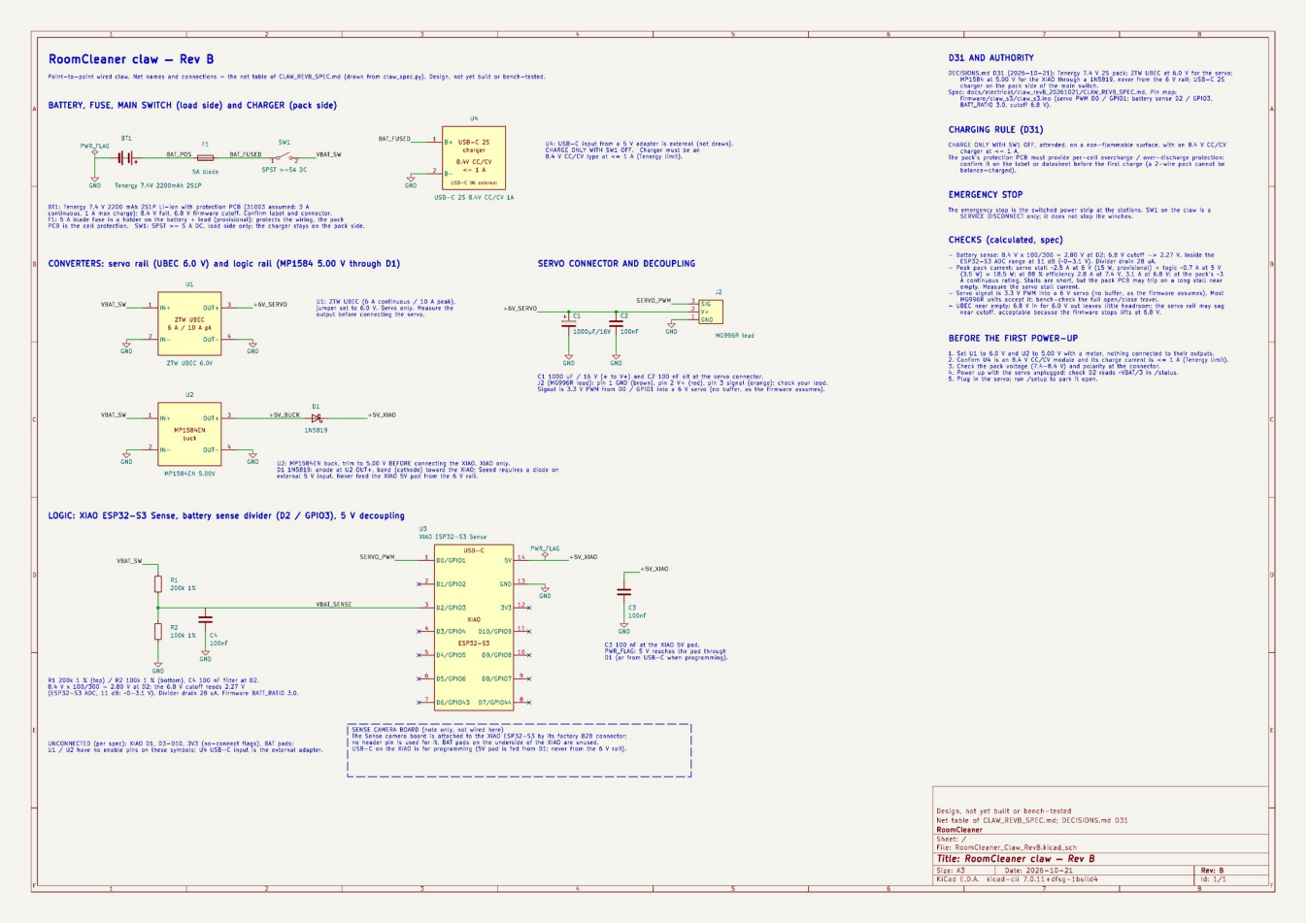
Phone app: home, live claw camera and chat

- Start up: self-check, take up slack, calibrate
- Find: scan from above, detect garments, choose the nearest
- Pick: hover until still, grip, deliver; the camera confirms each step
- Every move waits for approval on the phone; PAUSE holds position

Control Logic: Generation 1 vs. Final

	Generation 1	Final
Where is the claw?	Assumed from the commanded cable lengths	Measured by the claw camera; refit at every startup
Room geometry	Anchors from a tape measure	Anchors and cable lengths fitted, with error bars
Motor control	One central board	A controller per station, coordinated by the laptop
Finding laundry	Ceiling camera	Claw camera
Did the grab work?	Assumed	Camera checks pickup and delivery
Faults	Emergency stop only	A stop keeps torque; nothing restarts by itself

Claw Electrical



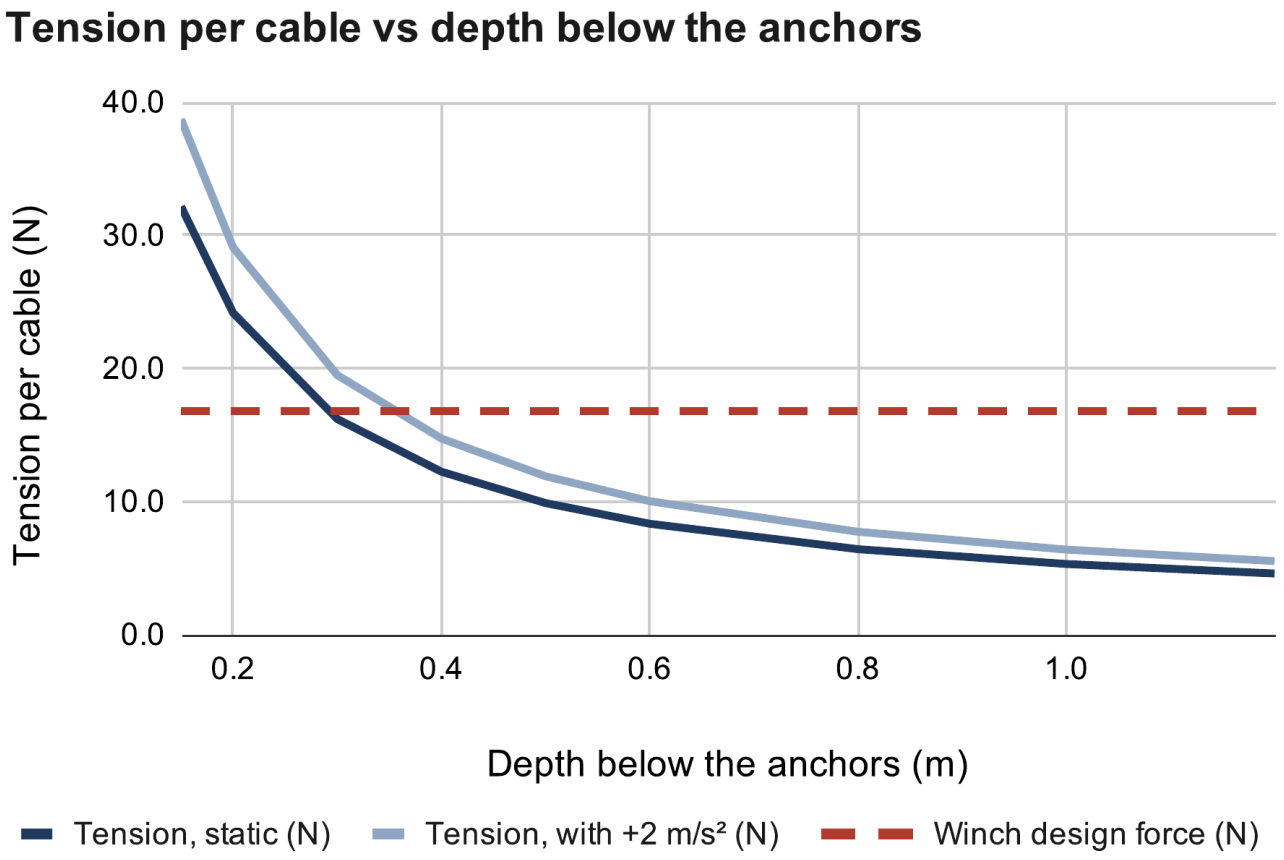
Claw schematic

- Two-cell lithium-ion pack: 7.4 V with a 5 A fuse
- 6 V servo rail and a separate 5 V logic rail
- Peak 2.8–3.1 A against a 3 A pack rating

Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
CNC	Computer numerical control
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988	Stepper-motor driver chip (Allegro)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress divided by actual stress

Calculations and Sources



Source: RoomCleaner engineering calculations, “Cable tension” (calculated)

- All charts come from my engineering calculation sheet
- Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
- Allegro MicroSystems, A4988 datasheet

RoomCleaner engineering calculations (Google Sheets)